STA 445 S26 Assignment 5

2026-03-18

Problem 1

Convert the following to date or date/time objects.

a. September 13, 2010.

```
mdy("September 13, 2010")
```

```
## [1] "2010-09-13"
```

b. Sept 13, 2010.

```
mdy("09-13-2010")
```

```
## [1] "2010-09-13"
```

c. Sep 13, 2010.

```
mdy("Sep 13, 2010")
```

```
## [1] "2010-09-13"
```

d. S 13, 2010. Comment on the month abbreviation needs.

```
mdy("Sep 13, 2010")
```

```
## [1] "2010-09-13"
```

The month needs to have a 3 letter abbreviation or the whole thing needs to be spelled out.

e. 07-Dec-1941.

```
mdy("12-07-1941")
```

```
## [1] "1941-12-07"
```

f. 1-5-1998. Comment on why you might be wrong.

```r
mdy("1-5-1998")
```

```
## [1] "1998-01-05"
```

The year or month could be 01 or it could be 05, so either January or March with the date being the 5th of January or the 1st of March

g. 21-5-1998. Comment on why you know you are correct.

```r
mdy("May 21, 1998")
```

```
## [1] "1998-05-21"
```

Since there is no month 21, I know this orientation has to be correct.

h. 2020-May-5 10:30 am

```r
mdy_hms("May 5 2020 10:30:00")
```

```
## [1] "2020-05-05 10:30:00 UTC"
```

i. 2020-May-5 10:30 am PDT (ex Seattle)

```r
#OlsonNames()
```

```r
ymd_hms("2020-05-05 10:30:00", tz="America/Los_Angeles")
```

```
## [1] "2020-05-05 10:30:00 PDT"
```

j. 2020-May-5 10:30 am AST (ex Puerto Rico)

```r
ymd_hms("2020-05-05 10:30:00", tz="America/Puerto_Rico")
```

```
## [1] "2020-05-05 10:30:00 AST"
```

Problem 2

Using just your date of birth (ex Sep 7, 1998) and today's date calculate the following:

```r
birth <- mdy("10-27-2004")
today <- mdy("3-16-2026")
```

a. Calculate the date of your 64th birthday.

```r
birth + years(64)
```

```
## [1] "2068-10-27"
```

b. Calculate your current age (in years).

```r
(age <- as.period(birth %--% today, unit = "years"))
```

```
## [1] "21y 4m 17d 0H 0M 0S"
```

c. Using your result in part (b), calculate the date of your next birthday.

d. The number of *days* until your next birthday.

```r
(next_birthday <- as.period((today%--%(birth+years(22))),unit= "days"))
```

```
## [1] "225d 0H 0M 0S"
```

e. The number of *months* and *days* until your next birthday.

```r
(next_birthday2 <- as.period((today%--%(birth+years(22)))))
```

```
## [1] "7m 11d 0H 0M 0S"
```

Problem 3

Suppose you have arranged for a phone call to be at 3 pm on May 8, 2015 at Arizona time. However, the recipient will be in Auckland, NZ. What time will it be there?

```r
Arizona <- ymd_hms("2015-05-08 15:00:00", tz = "US/Arizona")
NZ <- with_tz(Arizona, tz = "Pacific/Auckland")
NZ
```

```
## [1] "2015-05-09 10:00:00 NZST"
```

Problem 4

It turns out there is some interesting periodicity regarding the number of births on particular days of the year.

a. Using the `mosaicData` package, load the data set `Births78` which records the number of children born on each day in the United States in 1978. Because this problem is intended to show how to calculate the information using the `date`, remove all the columns *except* `date` and `births`.

```r
data("Births78")
(new <- Births78[, c('date','births')])
```

```
##           date births
## 1   1978-01-01   7701
## 2   1978-01-02   7527
## 3   1978-01-03   8825
## 4   1978-01-04   8859
## 5   1978-01-05   9043
## 6   1978-01-06   9208
## 7   1978-01-07   8084
## 8   1978-01-08   7611
## 9   1978-01-09   9172
## 10  1978-01-10   9089
## 11  1978-01-11   9210
## 12  1978-01-12   9259
## 13  1978-01-13   9138
## 14  1978-01-14   8299
## 15  1978-01-15   7771
## 16  1978-01-16   9458
## 17  1978-01-17   9339
## 18  1978-01-18   9120
## 19  1978-01-19   9226
## 20  1978-01-20   9305
## 21  1978-01-21   7954
## 22  1978-01-22   7560
## 23  1978-01-23   9252
## 24  1978-01-24   9416
## 25  1978-01-25   9090
## 26  1978-01-26   9387
## 27  1978-01-27   8983
## 28  1978-01-28   7946
## 29  1978-01-29   7527
## 30  1978-01-30   9184
## 31  1978-01-31   9152
## 32  1978-02-01   9159
## 33  1978-02-02   9218
## 34  1978-02-03   9167
## 35  1978-02-04   8065
## 36  1978-02-05   7804
## 37  1978-02-06   9225
## 38  1978-02-07   9328
## 39  1978-02-08   9139
## 40  1978-02-09   9247
## 41  1978-02-10   9527
## 42  1978-02-11   8144
## 43  1978-02-12   7950
## 44  1978-02-13   8966
## 45  1978-02-14   9859
```

```
## 46  1978-02-15   9285
## 47  1978-02-16   9103
## 48  1978-02-17   9238
## 49  1978-02-18   8167
## 50  1978-02-19   7695
## 51  1978-02-20   9021
## 52  1978-02-21   9252
## 53  1978-02-22   9335
## 54  1978-02-23   9268
## 55  1978-02-24   9552
## 56  1978-02-25   8313
## 57  1978-02-26   7881
## 58  1978-02-27   9262
## 59  1978-02-28   9705
## 60  1978-03-01   9132
## 61  1978-03-02   9304
## 62  1978-03-03   9431
## 63  1978-03-04   8008
## 64  1978-03-05   7791
## 65  1978-03-06   9294
## 66  1978-03-07   9573
## 67  1978-03-08   9212
## 68  1978-03-09   9218
## 69  1978-03-10   9583
## 70  1978-03-11   8144
## 71  1978-03-12   7870
## 72  1978-03-13   9022
## 73  1978-03-14   9525
## 74  1978-03-15   9284
## 75  1978-03-16   9327
## 76  1978-03-17   9480
## 77  1978-03-18   7965
## 78  1978-03-19   7729
## 79  1978-03-20   9135
## 80  1978-03-21   9663
## 81  1978-03-22   9307
## 82  1978-03-23   9159
## 83  1978-03-24   9157
## 84  1978-03-25   7874
## 85  1978-03-26   7589
## 86  1978-03-27   9100
## 87  1978-03-28   9293
## 88  1978-03-29   9195
## 89  1978-03-30   8902
## 90  1978-03-31   9318
## 91  1978-04-01   8069
## 92  1978-04-02   7691
## 93  1978-04-03   9114
## 94  1978-04-04   9439
## 95  1978-04-05   8852
```

```
## 96  1978-04-06   8969
## 97  1978-04-07   9077
## 98  1978-04-08   7890
## 99  1978-04-09   7445
## 100 1978-04-10   8870
## 101 1978-04-11   9023
## 102 1978-04-12   8606
## 103 1978-04-13   8724
## 104 1978-04-14   9012
## 105 1978-04-15   7527
## 106 1978-04-16   7193
## 107 1978-04-17   8702
## 108 1978-04-18   9205
## 109 1978-04-19   8720
## 110 1978-04-20   8582
## 111 1978-04-21   8892
## 112 1978-04-22   7787
## 113 1978-04-23   7304
## 114 1978-04-24   9017
## 115 1978-04-25   9077
## 116 1978-04-26   9019
## 117 1978-04-27   8839
## 118 1978-04-28   9047
## 119 1978-04-29   7750
## 120 1978-04-30   7135
## 121 1978-05-01   8900
## 122 1978-05-02   9422
## 123 1978-05-03   9051
## 124 1978-05-04   8672
## 125 1978-05-05   9101
## 126 1978-05-06   7718
## 127 1978-05-07   7388
## 128 1978-05-08   8987
## 129 1978-05-09   9307
## 130 1978-05-10   9273
## 131 1978-05-11   8903
## 132 1978-05-12   8975
## 133 1978-05-13   7762
## 134 1978-05-14   7382
## 135 1978-05-15   9195
## 136 1978-05-16   9200
## 137 1978-05-17   8913
## 138 1978-05-18   9044
## 139 1978-05-19   9000
## 140 1978-05-20   8064
## 141 1978-05-21   7570
## 142 1978-05-22   9089
## 143 1978-05-23   9210
## 144 1978-05-24   9196
## 145 1978-05-25   9180
```

```
## 146 1978-05-26   9514
## 147 1978-05-27   8005
## 148 1978-05-28   7781
## 149 1978-05-29   7780
## 150 1978-05-30   9630
## 151 1978-05-31   9600
## 152 1978-06-01   9435
## 153 1978-06-02   9303
## 154 1978-06-03   7971
## 155 1978-06-04   7399
## 156 1978-06-05   9127
## 157 1978-06-06   9606
## 158 1978-06-07   9328
## 159 1978-06-08   9075
## 160 1978-06-09   9362
## 161 1978-06-10   8040
## 162 1978-06-11   7581
## 163 1978-06-12   9201
## 164 1978-06-13   9264
## 165 1978-06-14   9216
## 166 1978-06-15   9175
## 167 1978-06-16   9350
## 168 1978-06-17   8233
## 169 1978-06-18   7777
## 170 1978-06-19   9543
## 171 1978-06-20   9672
## 172 1978-06-21   9266
## 173 1978-06-22   9405
## 174 1978-06-23   9598
## 175 1978-06-24   8122
## 176 1978-06-25   8091
## 177 1978-06-26   9348
## 178 1978-06-27   9857
## 179 1978-06-28   9701
## 180 1978-06-29   9630
## 181 1978-06-30  10080
## 182 1978-07-01   8209
## 183 1978-07-02   7976
## 184 1978-07-03   9284
## 185 1978-07-04   8433
## 186 1978-07-05   9675
## 187 1978-07-06  10184
## 188 1978-07-07  10241
## 189 1978-07-08   8773
## 190 1978-07-09   8102
## 191 1978-07-10   9877
## 192 1978-07-11   9852
## 193 1978-07-12   9705
## 194 1978-07-13   9984
## 195 1978-07-14  10438
```

```
## 196 1978-07-15   8859
## 197 1978-07-16   8416
## 198 1978-07-17  10026
## 199 1978-07-18  10357
## 200 1978-07-19  10015
## 201 1978-07-20  10386
## 202 1978-07-21  10332
## 203 1978-07-22   9062
## 204 1978-07-23   8563
## 205 1978-07-24   9960
## 206 1978-07-25  10349
## 207 1978-07-26  10091
## 208 1978-07-27  10192
## 209 1978-07-28  10307
## 210 1978-07-29   8677
## 211 1978-07-30   8486
## 212 1978-07-31   9890
## 213 1978-08-01  10145
## 214 1978-08-02   9824
## 215 1978-08-03  10128
## 216 1978-08-04  10051
## 217 1978-08-05   8738
## 218 1978-08-06   8442
## 219 1978-08-07  10206
## 220 1978-08-08  10442
## 221 1978-08-09  10142
## 222 1978-08-10  10284
## 223 1978-08-11  10162
## 224 1978-08-12   8951
## 225 1978-08-13   8532
## 226 1978-08-14  10127
## 227 1978-08-15  10502
## 228 1978-08-16  10053
## 229 1978-08-17  10377
## 230 1978-08-18  10355
## 231 1978-08-19   8904
## 232 1978-08-20   8477
## 233 1978-08-21   9967
## 234 1978-08-22  10229
## 235 1978-08-23   9900
## 236 1978-08-24  10152
## 237 1978-08-25  10173
## 238 1978-08-26   8782
## 239 1978-08-27   8453
## 240 1978-08-28   9998
## 241 1978-08-29  10387
## 242 1978-08-30  10063
## 243 1978-08-31   9849
## 244 1978-09-01  10114
## 245 1978-09-02   8580
```

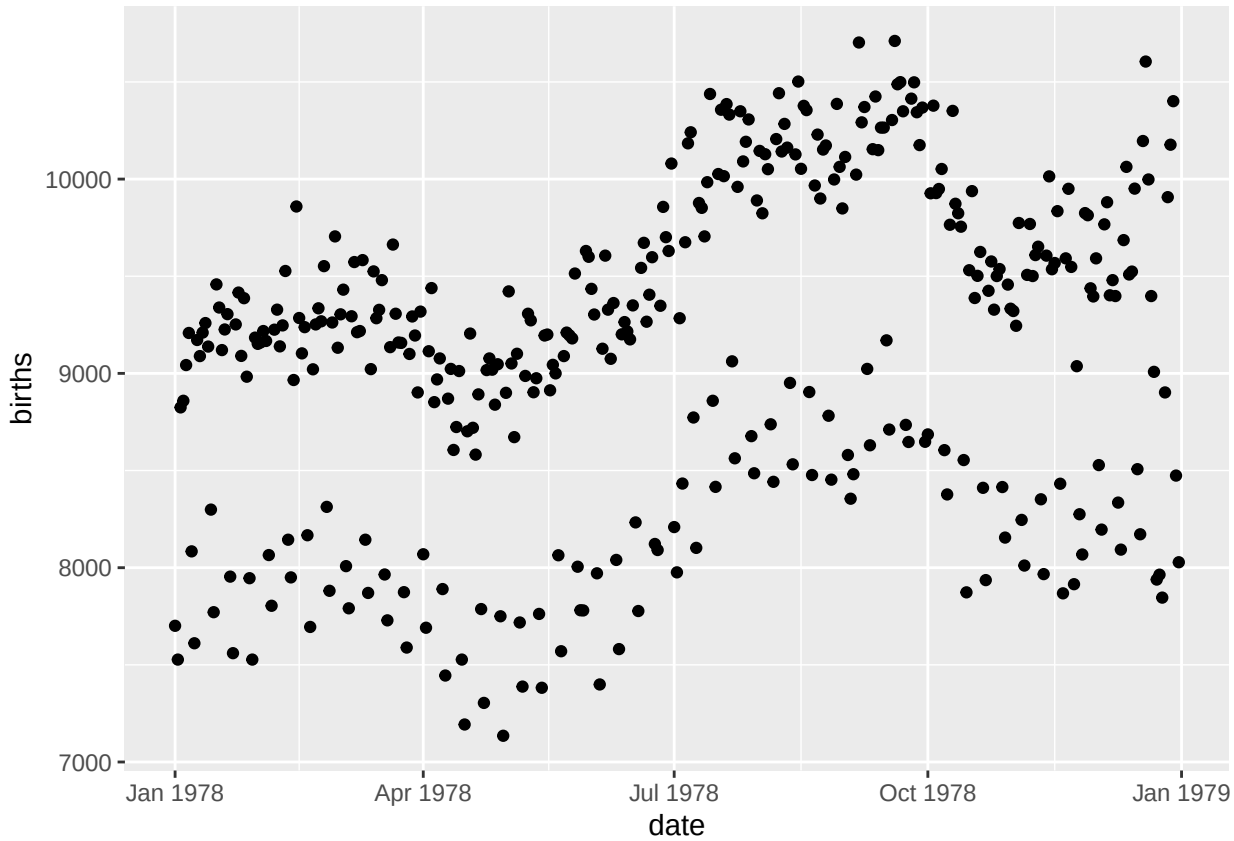
```
## 246 1978-09-03   8355
## 247 1978-09-04   8481
## 248 1978-09-05  10023
## 249 1978-09-06  10703
## 250 1978-09-07  10292
## 251 1978-09-08  10371
## 252 1978-09-09   9023
## 253 1978-09-10   8630
## 254 1978-09-11  10154
## 255 1978-09-12  10425
## 256 1978-09-13  10149
## 257 1978-09-14  10265
## 258 1978-09-15  10265
## 259 1978-09-16   9170
## 260 1978-09-17   8711
## 261 1978-09-18  10304
## 262 1978-09-19  10711
## 263 1978-09-20  10488
## 264 1978-09-21  10499
## 265 1978-09-22  10349
## 266 1978-09-23   8735
## 267 1978-09-24   8647
## 268 1978-09-25  10414
## 269 1978-09-26  10498
## 270 1978-09-27  10344
## 271 1978-09-28  10175
## 272 1978-09-29  10368
## 273 1978-09-30   8648
## 274 1978-10-01   8686
## 275 1978-10-02   9927
## 276 1978-10-03  10378
## 277 1978-10-04   9928
## 278 1978-10-05   9949
## 279 1978-10-06  10052
## 280 1978-10-07   8605
## 281 1978-10-08   8377
## 282 1978-10-09   9765
## 283 1978-10-10  10351
## 284 1978-10-11   9873
## 285 1978-10-12   9824
## 286 1978-10-13   9755
## 287 1978-10-14   8554
## 288 1978-10-15   7873
## 289 1978-10-16   9531
## 290 1978-10-17   9938
## 291 1978-10-18   9388
## 292 1978-10-19   9502
## 293 1978-10-20   9625
## 294 1978-10-21   8411
## 295 1978-10-22   7936
```

```
## 296 1978-10-23   9425
## 297 1978-10-24   9576
## 298 1978-10-25   9328
## 299 1978-10-26   9501
## 300 1978-10-27   9537
## 301 1978-10-28   8415
## 302 1978-10-29   8155
## 303 1978-10-30   9457
## 304 1978-10-31   9333
## 305 1978-11-01   9321
## 306 1978-11-02   9245
## 307 1978-11-03   9774
## 308 1978-11-04   8246
## 309 1978-11-05   8011
## 310 1978-11-06   9507
## 311 1978-11-07   9769
## 312 1978-11-08   9501
## 313 1978-11-09   9609
## 314 1978-11-10   9652
## 315 1978-11-11   8352
## 316 1978-11-12   7967
## 317 1978-11-13   9606
## 318 1978-11-14  10014
## 319 1978-11-15   9536
## 320 1978-11-16   9568
## 321 1978-11-17   9835
## 322 1978-11-18   8432
## 323 1978-11-19   7868
## 324 1978-11-20   9592
## 325 1978-11-21   9950
## 326 1978-11-22   9548
## 327 1978-11-23   7915
## 328 1978-11-24   9037
## 329 1978-11-25   8275
## 330 1978-11-26   8068
## 331 1978-11-27   9825
## 332 1978-11-28   9814
## 333 1978-11-29   9438
## 334 1978-11-30   9396
## 335 1978-12-01   9592
## 336 1978-12-02   8528
## 337 1978-12-03   8196
## 338 1978-12-04   9767
## 339 1978-12-05   9881
## 340 1978-12-06   9402
## 341 1978-12-07   9480
## 342 1978-12-08   9398
## 343 1978-12-09   8335
## 344 1978-12-10   8093
## 345 1978-12-11   9686
```

```
## 346 1978-12-12  10063
## 347 1978-12-13   9509
## 348 1978-12-14   9524
## 349 1978-12-15   9951
## 350 1978-12-16   8507
## 351 1978-12-17   8172
## 352 1978-12-18  10196
## 353 1978-12-19  10605
## 354 1978-12-20   9998
## 355 1978-12-21   9398
## 356 1978-12-22   9008
## 357 1978-12-23   7939
## 358 1978-12-24   7964
## 359 1978-12-25   7846
## 360 1978-12-26   8902
## 361 1978-12-27   9907
## 362 1978-12-28  10177
## 363 1978-12-29  10401
## 364 1978-12-30   8474
## 365 1978-12-31   8028
```

b. Graph the number of `births` vs the `date` with date on the x-axis. What stands out to you? Why do you think we have this trend?

```r
ggplot(new,aes(x = date,y = births)) +
  geom_point()
```



There are two separate "groups" on how many births happen on certain days. I believe that people have less kids on a certain day which is why this happens.

c. To test your assumption, we need to figure out the what day of the week each observation is. Use `dplyr::mutate` to add a new column named `dow` that is the day of the week (Monday, Tuesday, etc). This calculation will involve some function in the `lubridate` package and the `date` column.

```
new_dow <- new %>%
  mutate(dow = wday(date, label = TRUE, abbr = TRUE))
new_dow
```

```
##           date births dow
## 1   1978-01-01   7701 Sun
## 2   1978-01-02   7527 Mon
## 3   1978-01-03   8825 Tue
## 4   1978-01-04   8859 Wed
## 5   1978-01-05   9043 Thu
## 6   1978-01-06   9208 Fri
## 7   1978-01-07   8084 Sat
## 8   1978-01-08   7611 Sun
## 9   1978-01-09   9172 Mon
## 10  1978-01-10   9089 Tue
## 11  1978-01-11   9210 Wed
## 12  1978-01-12   9259 Thu
## 13  1978-01-13   9138 Fri
```

```
## 14  1978-01-14   8299 Sat
## 15  1978-01-15   7771 Sun
## 16  1978-01-16   9458 Mon
## 17  1978-01-17   9339 Tue
## 18  1978-01-18   9120 Wed
## 19  1978-01-19   9226 Thu
## 20  1978-01-20   9305 Fri
## 21  1978-01-21   7954 Sat
## 22  1978-01-22   7560 Sun
## 23  1978-01-23   9252 Mon
## 24  1978-01-24   9416 Tue
## 25  1978-01-25   9090 Wed
## 26  1978-01-26   9387 Thu
## 27  1978-01-27   8983 Fri
## 28  1978-01-28   7946 Sat
## 29  1978-01-29   7527 Sun
## 30  1978-01-30   9184 Mon
## 31  1978-01-31   9152 Tue
## 32  1978-02-01   9159 Wed
## 33  1978-02-02   9218 Thu
## 34  1978-02-03   9167 Fri
## 35  1978-02-04   8065 Sat
## 36  1978-02-05   7804 Sun
## 37  1978-02-06   9225 Mon
## 38  1978-02-07   9328 Tue
## 39  1978-02-08   9139 Wed
## 40  1978-02-09   9247 Thu
## 41  1978-02-10   9527 Fri
## 42  1978-02-11   8144 Sat
## 43  1978-02-12   7950 Sun
## 44  1978-02-13   8966 Mon
## 45  1978-02-14   9859 Tue
## 46  1978-02-15   9285 Wed
## 47  1978-02-16   9103 Thu
## 48  1978-02-17   9238 Fri
## 49  1978-02-18   8167 Sat
## 50  1978-02-19   7695 Sun
## 51  1978-02-20   9021 Mon
## 52  1978-02-21   9252 Tue
## 53  1978-02-22   9335 Wed
## 54  1978-02-23   9268 Thu
## 55  1978-02-24   9552 Fri
## 56  1978-02-25   8313 Sat
## 57  1978-02-26   7881 Sun
## 58  1978-02-27   9262 Mon
## 59  1978-02-28   9705 Tue
## 60  1978-03-01   9132 Wed
## 61  1978-03-02   9304 Thu
## 62  1978-03-03   9431 Fri
## 63  1978-03-04   8008 Sat
```

```
## 64  1978-03-05   7791 Sun
## 65  1978-03-06   9294 Mon
## 66  1978-03-07   9573 Tue
## 67  1978-03-08   9212 Wed
## 68  1978-03-09   9218 Thu
## 69  1978-03-10   9583 Fri
## 70  1978-03-11   8144 Sat
## 71  1978-03-12   7870 Sun
## 72  1978-03-13   9022 Mon
## 73  1978-03-14   9525 Tue
## 74  1978-03-15   9284 Wed
## 75  1978-03-16   9327 Thu
## 76  1978-03-17   9480 Fri
## 77  1978-03-18   7965 Sat
## 78  1978-03-19   7729 Sun
## 79  1978-03-20   9135 Mon
## 80  1978-03-21   9663 Tue
## 81  1978-03-22   9307 Wed
## 82  1978-03-23   9159 Thu
## 83  1978-03-24   9157 Fri
## 84  1978-03-25   7874 Sat
## 85  1978-03-26   7589 Sun
## 86  1978-03-27   9100 Mon
## 87  1978-03-28   9293 Tue
## 88  1978-03-29   9195 Wed
## 89  1978-03-30   8902 Thu
## 90  1978-03-31   9318 Fri
## 91  1978-04-01   8069 Sat
## 92  1978-04-02   7691 Sun
## 93  1978-04-03   9114 Mon
## 94  1978-04-04   9439 Tue
## 95  1978-04-05   8852 Wed
## 96  1978-04-06   8969 Thu
## 97  1978-04-07   9077 Fri
## 98  1978-04-08   7890 Sat
## 99  1978-04-09   7445 Sun
## 100 1978-04-10   8870 Mon
## 101 1978-04-11   9023 Tue
## 102 1978-04-12   8606 Wed
## 103 1978-04-13   8724 Thu
## 104 1978-04-14   9012 Fri
## 105 1978-04-15   7527 Sat
## 106 1978-04-16   7193 Sun
## 107 1978-04-17   8702 Mon
## 108 1978-04-18   9205 Tue
## 109 1978-04-19   8720 Wed
## 110 1978-04-20   8582 Thu
## 111 1978-04-21   8892 Fri
## 112 1978-04-22   7787 Sat
## 113 1978-04-23   7304 Sun
```

```
## 114 1978-04-24   9017 Mon
## 115 1978-04-25   9077 Tue
## 116 1978-04-26   9019 Wed
## 117 1978-04-27   8839 Thu
## 118 1978-04-28   9047 Fri
## 119 1978-04-29   7750 Sat
## 120 1978-04-30   7135 Sun
## 121 1978-05-01   8900 Mon
## 122 1978-05-02   9422 Tue
## 123 1978-05-03   9051 Wed
## 124 1978-05-04   8672 Thu
## 125 1978-05-05   9101 Fri
## 126 1978-05-06   7718 Sat
## 127 1978-05-07   7388 Sun
## 128 1978-05-08   8987 Mon
## 129 1978-05-09   9307 Tue
## 130 1978-05-10   9273 Wed
## 131 1978-05-11   8903 Thu
## 132 1978-05-12   8975 Fri
## 133 1978-05-13   7762 Sat
## 134 1978-05-14   7382 Sun
## 135 1978-05-15   9195 Mon
## 136 1978-05-16   9200 Tue
## 137 1978-05-17   8913 Wed
## 138 1978-05-18   9044 Thu
## 139 1978-05-19   9000 Fri
## 140 1978-05-20   8064 Sat
## 141 1978-05-21   7570 Sun
## 142 1978-05-22   9089 Mon
## 143 1978-05-23   9210 Tue
## 144 1978-05-24   9196 Wed
## 145 1978-05-25   9180 Thu
## 146 1978-05-26   9514 Fri
## 147 1978-05-27   8005 Sat
## 148 1978-05-28   7781 Sun
## 149 1978-05-29   7780 Mon
## 150 1978-05-30   9630 Tue
## 151 1978-05-31   9600 Wed
## 152 1978-06-01   9435 Thu
## 153 1978-06-02   9303 Fri
## 154 1978-06-03   7971 Sat
## 155 1978-06-04   7399 Sun
## 156 1978-06-05   9127 Mon
## 157 1978-06-06   9606 Tue
## 158 1978-06-07   9328 Wed
## 159 1978-06-08   9075 Thu
## 160 1978-06-09   9362 Fri
## 161 1978-06-10   8040 Sat
## 162 1978-06-11   7581 Sun
## 163 1978-06-12   9201 Mon
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## 164 1978-06-13   9264 Tue
## 165 1978-06-14   9216 Wed
## 166 1978-06-15   9175 Thu
## 167 1978-06-16   9350 Fri
## 168 1978-06-17   8233 Sat
## 169 1978-06-18   7777 Sun
## 170 1978-06-19   9543 Mon
## 171 1978-06-20   9672 Tue
## 172 1978-06-21   9266 Wed
## 173 1978-06-22   9405 Thu
## 174 1978-06-23   9598 Fri
## 175 1978-06-24   8122 Sat
## 176 1978-06-25   8091 Sun
## 177 1978-06-26   9348 Mon
## 178 1978-06-27   9857 Tue
## 179 1978-06-28   9701 Wed
## 180 1978-06-29   9630 Thu
## 181 1978-06-30  10080 Fri
## 182 1978-07-01   8209 Sat
## 183 1978-07-02   7976 Sun
## 184 1978-07-03   9284 Mon
## 185 1978-07-04   8433 Tue
## 186 1978-07-05   9675 Wed
## 187 1978-07-06  10184 Thu
## 188 1978-07-07  10241 Fri
## 189 1978-07-08   8773 Sat
## 190 1978-07-09   8102 Sun
## 191 1978-07-10   9877 Mon
## 192 1978-07-11   9852 Tue
## 193 1978-07-12   9705 Wed
## 194 1978-07-13   9984 Thu
## 195 1978-07-14  10438 Fri
## 196 1978-07-15   8859 Sat
## 197 1978-07-16   8416 Sun
## 198 1978-07-17  10026 Mon
## 199 1978-07-18  10357 Tue
## 200 1978-07-19  10015 Wed
## 201 1978-07-20  10386 Thu
## 202 1978-07-21  10332 Fri
## 203 1978-07-22   9062 Sat
## 204 1978-07-23   8563 Sun
## 205 1978-07-24   9960 Mon
## 206 1978-07-25  10349 Tue
## 207 1978-07-26  10091 Wed
## 208 1978-07-27  10192 Thu
## 209 1978-07-28  10307 Fri
## 210 1978-07-29   8677 Sat
## 211 1978-07-30   8486 Sun
## 212 1978-07-31   9890 Mon
## 213 1978-08-01  10145 Tue
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## 214 1978-08-02   9824 Wed
## 215 1978-08-03  10128 Thu
## 216 1978-08-04  10051 Fri
## 217 1978-08-05   8738 Sat
## 218 1978-08-06   8442 Sun
## 219 1978-08-07  10206 Mon
## 220 1978-08-08  10442 Tue
## 221 1978-08-09  10142 Wed
## 222 1978-08-10  10284 Thu
## 223 1978-08-11  10162 Fri
## 224 1978-08-12   8951 Sat
## 225 1978-08-13   8532 Sun
## 226 1978-08-14  10127 Mon
## 227 1978-08-15  10502 Tue
## 228 1978-08-16  10053 Wed
## 229 1978-08-17  10377 Thu
## 230 1978-08-18  10355 Fri
## 231 1978-08-19   8904 Sat
## 232 1978-08-20   8477 Sun
## 233 1978-08-21   9967 Mon
## 234 1978-08-22  10229 Tue
## 235 1978-08-23   9900 Wed
## 236 1978-08-24  10152 Thu
## 237 1978-08-25  10173 Fri
## 238 1978-08-26   8782 Sat
## 239 1978-08-27   8453 Sun
## 240 1978-08-28   9998 Mon
## 241 1978-08-29  10387 Tue
## 242 1978-08-30  10063 Wed
## 243 1978-08-31   9849 Thu
## 244 1978-09-01  10114 Fri
## 245 1978-09-02   8580 Sat
## 246 1978-09-03   8355 Sun
## 247 1978-09-04   8481 Mon
## 248 1978-09-05  10023 Tue
## 249 1978-09-06  10703 Wed
## 250 1978-09-07  10292 Thu
## 251 1978-09-08  10371 Fri
## 252 1978-09-09   9023 Sat
## 253 1978-09-10   8630 Sun
## 254 1978-09-11  10154 Mon
## 255 1978-09-12  10425 Tue
## 256 1978-09-13  10149 Wed
## 257 1978-09-14  10265 Thu
## 258 1978-09-15  10265 Fri
## 259 1978-09-16   9170 Sat
## 260 1978-09-17   8711 Sun
## 261 1978-09-18  10304 Mon
## 262 1978-09-19  10711 Tue
## 263 1978-09-20  10488 Wed
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## 264 1978-09-21  10499 Thu
## 265 1978-09-22  10349 Fri
## 266 1978-09-23   8735 Sat
## 267 1978-09-24   8647 Sun
## 268 1978-09-25  10414 Mon
## 269 1978-09-26  10498 Tue
## 270 1978-09-27  10344 Wed
## 271 1978-09-28  10175 Thu
## 272 1978-09-29  10368 Fri
## 273 1978-09-30   8648 Sat
## 274 1978-10-01   8686 Sun
## 275 1978-10-02   9927 Mon
## 276 1978-10-03  10378 Tue
## 277 1978-10-04   9928 Wed
## 278 1978-10-05   9949 Thu
## 279 1978-10-06  10052 Fri
## 280 1978-10-07   8605 Sat
## 281 1978-10-08   8377 Sun
## 282 1978-10-09   9765 Mon
## 283 1978-10-10  10351 Tue
## 284 1978-10-11   9873 Wed
## 285 1978-10-12   9824 Thu
## 286 1978-10-13   9755 Fri
## 287 1978-10-14   8554 Sat
## 288 1978-10-15   7873 Sun
## 289 1978-10-16   9531 Mon
## 290 1978-10-17   9938 Tue
## 291 1978-10-18   9388 Wed
## 292 1978-10-19   9502 Thu
## 293 1978-10-20   9625 Fri
## 294 1978-10-21   8411 Sat
## 295 1978-10-22   7936 Sun
## 296 1978-10-23   9425 Mon
## 297 1978-10-24   9576 Tue
## 298 1978-10-25   9328 Wed
## 299 1978-10-26   9501 Thu
## 300 1978-10-27   9537 Fri
## 301 1978-10-28   8415 Sat
## 302 1978-10-29   8155 Sun
## 303 1978-10-30   9457 Mon
## 304 1978-10-31   9333 Tue
## 305 1978-11-01   9321 Wed
## 306 1978-11-02   9245 Thu
## 307 1978-11-03   9774 Fri
## 308 1978-11-04   8246 Sat
## 309 1978-11-05   8011 Sun
## 310 1978-11-06   9507 Mon
## 311 1978-11-07   9769 Tue
## 312 1978-11-08   9501 Wed
## 313 1978-11-09   9609 Thu
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## 314 1978-11-10   9652 Fri
## 315 1978-11-11   8352 Sat
## 316 1978-11-12   7967 Sun
## 317 1978-11-13   9606 Mon
## 318 1978-11-14  10014 Tue
## 319 1978-11-15   9536 Wed
## 320 1978-11-16   9568 Thu
## 321 1978-11-17   9835 Fri
## 322 1978-11-18   8432 Sat
## 323 1978-11-19   7868 Sun
## 324 1978-11-20   9592 Mon
## 325 1978-11-21   9950 Tue
## 326 1978-11-22   9548 Wed
## 327 1978-11-23   7915 Thu
## 328 1978-11-24   9037 Fri
## 329 1978-11-25   8275 Sat
## 330 1978-11-26   8068 Sun
## 331 1978-11-27   9825 Mon
## 332 1978-11-28   9814 Tue
## 333 1978-11-29   9438 Wed
## 334 1978-11-30   9396 Thu
## 335 1978-12-01   9592 Fri
## 336 1978-12-02   8528 Sat
## 337 1978-12-03   8196 Sun
## 338 1978-12-04   9767 Mon
## 339 1978-12-05   9881 Tue
## 340 1978-12-06   9402 Wed
## 341 1978-12-07   9480 Thu
## 342 1978-12-08   9398 Fri
## 343 1978-12-09   8335 Sat
## 344 1978-12-10   8093 Sun
## 345 1978-12-11   9686 Mon
## 346 1978-12-12  10063 Tue
## 347 1978-12-13   9509 Wed
## 348 1978-12-14   9524 Thu
## 349 1978-12-15   9951 Fri
## 350 1978-12-16   8507 Sat
## 351 1978-12-17   8172 Sun
## 352 1978-12-18  10196 Mon
## 353 1978-12-19  10605 Tue
## 354 1978-12-20   9998 Wed
## 355 1978-12-21   9398 Thu
## 356 1978-12-22   9008 Fri
## 357 1978-12-23   7939 Sat
## 358 1978-12-24   7964 Sun
## 359 1978-12-25   7846 Mon
## 360 1978-12-26   8902 Tue
## 361 1978-12-27   9907 Wed
## 362 1978-12-28  10177 Thu
## 363 1978-12-29  10401 Fri
```

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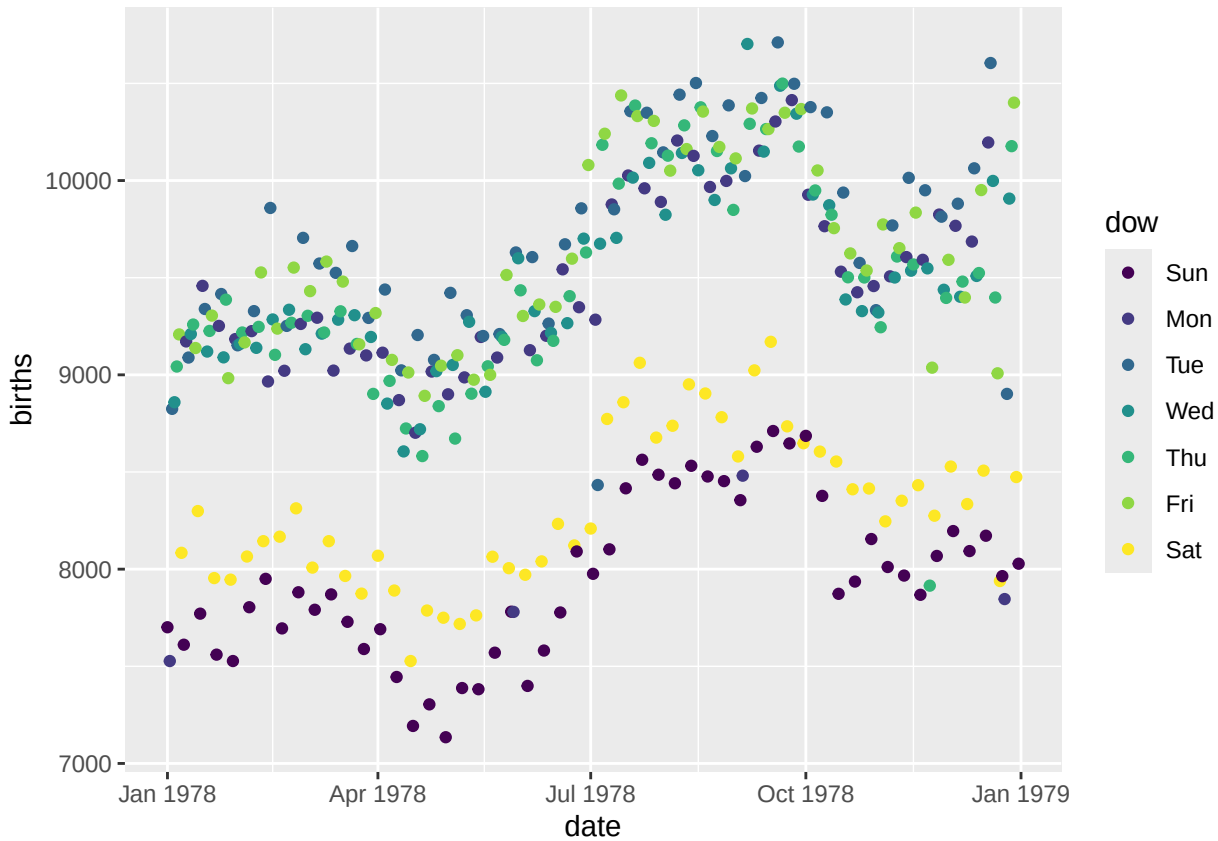
## 364 1978-12-30   8474 Sat
## 365 1978-12-31   8028 Sun
```

d. Plot the data with the point color being determined by the day of the week variable.

```
ggplot(new_dow,aes(x = date,y = births, color = dow)) +
  geom_point()
```